

Estimation Practice Problem

Statistics 7301

The BFC is a franchise fast-food restaurant specializing in burgers, fish sandwiches, and chicken sandwiches. Soft drinks and French fries are also available. The Planning Department of BFC reports that the distribution of daily sales for restaurants follows the normal distribution and that the population standard deviation is BDT 3,000. A sample of 40 showed the mean daily sales to be BDT 20,000.

Requirement:

- i. What does the population mean?
- ii. What does the best estimate of the population mean? What is this value called?
- iii. Develop a 99 percent confidence interval for the population mean.
- iv. Interpret the confidence interval.

- **Given:**
- Population standard deviation: $\sigma = 3000$
- Sample size: $n = 40$
- Sample mean: $\bar{x} = 20000$
- Distribution: Normal

$$Z = 2.576$$

Standard Error (SE)

$$SE = \frac{3000}{\sqrt{40}} = \frac{3000}{6.324} \approx 474.34$$

- **Step 1: Find $Z_{\alpha/2}$**
- For 99% confidence level:

$$Z_{\alpha/2} = 2.576$$

- **Step 2: Standard Error (SE)**

- $SE = \frac{3000}{\sqrt{40}} = \frac{3000}{6.324} \approx 474.34$

- **Step 3: Margin of Error (ME)**

- $ME = 2.576 \times 474.34 \approx 1222.7$

- **Step 4: Confidence Interval**

- $CI = 20000 \pm 1222.7$

- $CI = (18777.3, 21222.7)$

Example 8-1

A publishing company has just published a new college textbook. Before the company decides the price at which to sell this textbook, it wants to know the average price of all such textbooks in the market. The research department at the company took a sample of 25 comparable textbooks and collected information on their prices. This information produces a mean price of \$145 for this sample. It is known that the standard deviation of the prices of all such textbooks is \$35 and the population of such prices is normal.

Example 8-1

- (a) What is the point estimate of the mean price of all such textbooks?
- (b) Construct a 90% confidence interval for the mean price of all such college textbooks.

Example 8-1: Solution

a)

$n = 25$, $\bar{x} = \$145$, and $\sigma = \$35$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{35}{\sqrt{25}} = \$7.00$$

Point estimate of $\mu = \bar{x} = \$145$

Example 8-1: Solution

b)

Confidence level is 90% or .90. Here, the area in each tail of the normal distribution curve is $\alpha/2 = (1 - .90)/2 = .05$. Hence, $z = 1.65$.

$$\begin{aligned}\bar{X} \pm z\sigma_{\bar{x}} &= 145 \pm 1.65(7.00) = 145 \pm 11.55 \\ &= (145 - 11.55) \text{ to } (145 + 11.55) \\ &= \$133.45 \text{ to } \$156.55\end{aligned}$$

Example 8-1: Solution

We can say that we are 90% confident that the mean price of all such college textbooks is between \$133.45 and \$156.55.

- The owner of BD Egg Farm wants to estimate the mean number of eggs produced per chicken. A sample of 20 chickens shows they produced an average of 20 eggs per month with a standard deviation of 2 eggs per month.

Requirement:

- What is the value of the population mean? What is the best estimate of this value?
- Explain which distribution should be used for this estimation problem.
- For a 95 percent confidence interval, what is the value of test statistic as identified in (ii)?
- Develop the 95 percent confidence interval for the population mean.
- Would it be reasonable to conclude that the population mean is 21 eggs? What about 25 eggs?

Appropriate distribution

Since:

- Population standard deviation is **unknown**
 - Sample size is **small ($n < 30$)**
- 👉 Use **t-distribution**

(iii) Test statistic for 95% CI

- Degrees of freedom: $df = n - 1 = 19$

From t-table:

$$t_{0.025,19} \approx 2.093$$

(iv) 95% Confidence Interval

Formula:

$$\bar{x} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Step 1: Standard Error

$$SE = \frac{2}{\sqrt{20}} \approx 0.447$$

Step 2: Margin of Error

$$ME = 2.093 \times 0.447 \approx 0.935$$

Step 3: Confidence Interval

$$CI = 20 \pm 0.935$$
$$CI = (19.065, 20.935)$$

A market research consultant was hired to estimate the proportion of homemakers who associate the brand name of a laundry detergent with the container's shape and colour. The consultant randomly selected 1,400 homemakers. From the sample, 420 were able to identify the brand by name based only on the shape and colour of the container.

Requirement:

- Estimate the value of the population proportion.
- Develop a 99 percent confidence interval for the population proportion.
- Interpret your findings.

- **Given:**

$$n = 1400$$
$$x = 420$$

- **(i) Estimate population proportion**

$$\hat{p} = \frac{420}{1400} = 0.30$$

Estimated proportion = **0.30 (30%)**

- (ii) 99% Confidence Interval**

- Formula:

$$\hat{p} \pm Z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Step 1: $Z_{\alpha/2}$

- For 99% confidence:

$$Z = 2.576$$

- **Step 2: Standard Error**

$$SE = \sqrt{\frac{0.30 \times 0.70}{1400}} \approx 0.01225$$

- **Step 3: Margin of Error**

$$ME = 2.576 \times 0.01225 \approx 0.0316$$

- **Step 4: Confidence Interval**

$$CI = 0.30 \pm 0.0316$$

$$CI = (0.2684, 0.3316)$$

Interpretation

- We are **99% confident** that the true proportion of homemakers who identify the brand based on shape and colour lies between:

26.84% and 33.16%

This indicates that **roughly one-third or fewer** homemakers associate the brand with packaging characteristics.